

# **TRI - FACTOR BIOMETRIC EVM**

*A Project Report Submitted in Partial fulfilment of the Requirements For the  
Degree of Bachelor of Technology in  
Electronics and Communication Engineering by:*

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## DECLARATION

We declare that the project report titled “**TRI - FACTOR BIOMETRIC EVM**” submitted in Partial Fulfilment of the Requirements for the Degree of Bachelor of Technology.

In Electronics and Communication Engineering is a record of work carried out by us under the guidance of **Dr. SK Imtiaj** and has not formed the basis for the award of any other degree, in this or any other Institution or University.

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# CERTIFICATE

This is to certify that **Soumodeep Mete, Rupak Maiti, Shruti Saha and Banisa Pramanick** have carried out their final year project work entitled “**TRI - FACTOR BIOMETRIC EVM**” as a part of the curriculum for the B.Tech Degree in Electronics and Communication Engineering (ECE) under Maulana Abul Kalam Azad University of Technology for the academic session 2025 - 2026.

This project report is approved by the undersigned only for the purpose for which it is submitted. The candidates are entirely responsible for the statements, opinions and conclusions contained herein.

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# ABSTRACT

Elections are the foundation of democracy, but the foundation of elections rests on one crucial question: Is this person really who they claim to be? In most of today's Electronic Voting Machines (EVMs), that answer is provided by a Polling Officer looking at a photo ID – a process that is vulnerable to human error, fatigue, and corruption. In this paper we propose the Tri - Factor Biometric Electronic Voting Machine (TF-BEVM) replacing officer judgment with three automated, independent and sequential identity checks, which cannot be bypassed simultaneously at any single point of failure.

The system forces authentication in three required steps. The voter is first verified by scanning his/her fingerprint on an R307S optical sensor at the officer's desk through a SQLite database. Second, a USB webcam for physical presence check by a real eye blink - the Eye Aspect Ratio ( $EAR < 0.28$ ) from the 468-point facial landmark mesh of MediaPipe—prior to deep-learning facial recognition by the FaceNet512 model, matching a 512-dimensional embedding with the enrolled record. Third, the voter rescans his finger inside the private voting booth, thereby sealing the vote against substitution attacks.

The hardware is based on a Master-Slave architecture, with an Arduino Uno Q (Qualcomm quad-core Linux processor with a real-time STM32U585 microcontroller) as the Control Unit at the officer's desk and an Arduino Uno R3 governing the isolated Balloting Unit inside the booth. All the voter data is locally stored, and there is no reliance on the internet. So, the system can be deployed both in urban polling stations and also in remote rural booths. A Python-based hardware watchdog manages all critical peripherals and automatically pauses/resumes the session on any disconnect with no loss of session state.

The system was tested in a controlled pseudo-voting setting with 22 voters. No fraudulent double votes were found; all photograph-spoofing attempts failed at the liveness stage, and all adversarial impersonation attempts failed at the fingerprint or facial recognition stage. Actual voter throughput was 90.90%, while the average end-to-end authentication time was 25.1 seconds—well within the capacity of a normal polling day.

This work shows that fully automated, trustless voter authentication is possible on low-cost embedded hardware, permanently closing the human-override loophole that remains the weakest link in today's electoral infrastructure.

**Keywords** – Biometric EVM, Fingerprint Authentication, R307S Sensor, Facial Recognition, FaceNet512, Liveness Detection, MediaPipe, Eye Aspect Ratio, Anti-Spoofing, Master-Slave Architecture, Arduino Uno Q, Qualcomm QRB2210, STM32U585, Arduino Uno R3, Socat TCP-Serial Bridge, SQLite WAL Mode, Continuous Ballot Monitoring, Hardware Watchdog, Offline Voting, Trustless Authentication.

# CHAPTER 1

## INTRODUCTION

India conducts the largest democratic exercise on Earth, managing more than 900 million registered voters spread across megacities, villages, forests, and mountains [1]. The history of Indian elections, however, is inseparable from the history of electoral malpractice — from ballot-stuffing and proxy voting in the post-independence decades to the organised violence of booth-capturing that made entire polling stations unavailable to legitimate voters throughout the 1970s and 1980s [2]. The introduction of the Electronic Voting Machine (EVM) represented a landmark response to this crisis. Piloted during the 1982 Kerala state election at Paravur constituency and fully deployed across all constituencies by 2004, EVMs eliminated invalid ballot papers, reduced counting time from days to hours, and made large-scale ballot-box theft physically impossible [3]. The standard EVM — designed jointly by ECIL and BEL — pairs a Control Unit at the officer's desk with a Balloting Unit inside the voting compartment [4]. The Voter Verifiable Paper Audit Trail (VVPAT), introduced in 2013, further added a printed confirmation slip visible to the voter for seven seconds before falling into a sealed container [5]. These were genuine milestones. But neither innovation addressed the foundational vulnerability that remains in place today.

The moment a voter walks into a polling booth, their identity is verified the same way it was fifty years ago: a human officer glances at their face and compares it to a photograph on a card. This human-in-the-loop approach is the weakest link in the entire electoral chain. An officer can be tired, distracted, bribed, or coerced. The machine records only that a button was pressed — not who pressed it. Authorization to vote is granted by the Presiding Officer pushing a button on the Control Unit, a human decision that can be manipulated. In January 2024, the Supreme Court of India ruled that ballot papers in the Chandigarh Municipal Corporation Mayor election had been deliberately defaced by a returning officer, describing the act as a "murder of democracy" [36]. In the same period, the Court's landmark judgment in *Association for Democratic Reforms v. Election Commission of India* [6] further reinforced the imperative of electoral transparency, striking down anonymous political funding mechanisms on the grounds that voters have a constitutional right to know the sources of electoral finance. Voter impersonation incidents were documented across multiple state elections throughout 2023 and 2024 [7]. These are not edge cases — they are the predictable, recurring consequences of a system that places the entire burden of identity verification on human judgment.

The most common proposed remedy is to add a fingerprint scanner to the existing EVM. This is a step forward, but it carries a critical flaw. A large portion of India's electorate — farmers, construction workers, factory labourers — have worn, cracked, or chemically damaged fingerprints that scanners cannot reliably read, with failure rates exceeding 10% in manual-labour demographics [43][28]. The standard engineering response is an Officer Override button that lets the presiding officer bypass the biometric check for voters whose prints do not scan. This single mechanism destroys the entire security promise of the system. A corrupt officer can press it for any fraudulent voter. The moment an override exists, security reverts

to depending entirely on human integrity — precisely the problem biometrics were introduced to eliminate.

This paper presents the Tri - Factor Biometric Electronic Voting Machine (TF-BEVM), designed from the ground up to remove that override permanently. The system authenticates every voter through three sequential and independent checks: a fingerprint scan at the officer's desk, passive liveness detection combined with deep-learning facial recognition via the FaceNet512 model [23], and a second fingerprint scan inside the private voting booth to seal the vote against substitution attacks. No bribed officer, no stolen identity card, no printed photograph, and no physical substitution can simultaneously defeat all three layers. All voter data is stored locally with zero network dependency, ensuring identical performance whether deployed in South Mumbai or a remote Sundarbans polling booth. The system boots and operates entirely autonomously — no connected laptop, no manual commands, no internet connection required at any stage of a live election.

# CHAPTER 2

## LITERATURE REVIEW

### 2.1 The Evolution of EVMs in India

India's voting journey began with paper ballots that were fundamentally vulnerable to booth-capturing — where organised groups would forcibly stamp ballots en masse, disenfranchising entire constituencies. The introduction of first-generation EVMs, piloted in the 1982 Kerala state election and fully deployed by 2004, resolved the invalid-vote problem and compressed counting time from days to hours [3]. The VVPAT, introduced in 2013, added a physical transparency layer by printing a confirmation slip visible to the voter for seven seconds [5]. VVPAT-EVM discrepancy analysis across multiple state elections has further revealed statistical anomalies in the matching process [38], a series of controversies chronicled in detail by The Hindu editorial board [37]. However, every generation of EVM has shared one structural weakness: the authorization to vote is controlled by a human officer, not by verified identity [4]. The Presiding Officer's button on the Control Unit remains the single point of human trust in an otherwise mechanical system — and it is the point this work is designed to eliminate.

### 2.2 Fingerprint-Only Biometric EVMs (2018–2021)

The first wave of biometric EVM research focused on adding fingerprint scanners to existing architectures. Pathak and Shukla [46] demonstrated fingerprint-based EVMs using the R307 optical sensor family that successfully enforced a one-person-one-vote policy under laboratory conditions. Jain and Ross [10] established the theoretical basis for such unimodal systems, while Snelick et al. [11] showed empirically that single-modality systems are consistently outperformed by multi-modal alternatives, reporting up to 40% reduction in equal error rate when a second modality is added. In practice, fingerprint-only deployments struggled with a large segment of the Indian electorate — farmers, construction workers, and manual labourers — whose ridge patterns are degraded by occupational wear. Beyond failure-to-enrol, Yoon et al. [27] demonstrated that degraded or partial fingerprint templates also increase the risk of reconstruction attacks, further undermining the security guarantees of single-modality fingerprint systems. Wayman et al. [43] documented that such skin deformation reduces fingerprint matching accuracy by up to 25% compared to office-worker reference populations [43]. The universal fallback was the Officer Override button, which, as discussed, reintroduces the human trust problem that biometrics were introduced to solve.

### 2.3 Cloud and IoT Voting Systems (2021–2023)

The second wave shifted toward networked architectures. Reddy et al. [8] proposed a cloud-based smart voting system using the ESP32 microcontroller and a Firebase real-time database, enabling flexible remote deployment. Singh and Gupta [49] introduced a blockchain-based system for producing immutable vote records, providing cryptographic auditability. Hipp et al. [32] documented the database-layer foundations for such systems, showing that SQLite in Write-Ahead Logging mode provides the concurrent read-write

consistency required for multithreaded voting backends. While technically sophisticated, both cloud-dependent approaches share a critical deployment assumption: reliable internet connectivity. The Election Commission of India explicitly mandates that voting infrastructure must function autonomously in areas with intermittent or absent connectivity [9] — an assumption that fails for hundreds of thousands of booths across India’s forests, highlands, and coastal islands. TF-BEVM addresses this by storing all voter records and biometric embeddings locally, with zero network dependency at any stage of operation.

## 2.4 Facial Recognition and the Spoofing Gap (2022–2024)

The third wave introduced facial recognition as a second authentication factor. Sharma et al. [50] developed a face-recognition voting prototype using Haar Cascade and LBPH algorithms on a Raspberry Pi. Kumar and Zhang [12] combined face and fingerprint modalities with CNN-based matching, and Gad et al. [13] specifically evaluated multimodal fusion for electronic voting, reporting 99.3% accuracy on a database of 500 synthetic voters. Schroff et al. [23] introduced FaceNet, training a deep convolutional network to optimise embedding distances directly using a triplet loss function — the architecture that TF-BEVM deploys through the FaceNet512 model, achieving 99.63% accuracy on the Labeled Faces in the Wild benchmark. An alternative line of work, exemplified by Wang et al.’s Additive Margin Softmax [25], reformulates face verification as a margin-based classification problem rather than a metric-learning one; while competitive in accuracy, this approach was not adopted for TF-BEVM because the triplet-loss embedding space of FaceNet512 integrates more directly with the per-voter cosine-distance thresholding described in Section III-B.4. However, all of these systems shared one catastrophic vulnerability: they could be defeated by holding a printed photograph in front of the camera. This presentation attack is trivially simple to execute and devastating in a voting context. Ramachandra and Busch [17] confirmed that 2D facial recognition without liveness detection is fundamentally insecure for high-stakes authentication. Ramachandra and Busch [17] surveyed face anti-spoofing methods across four categories — texture-based, depth-based, motion-based, and challenge-response — and identified passive blink detection as the most practical low-cost defence for embedded deployments. Soukupová and Čech [15] formalised this through the Eye Aspect Ratio (EAR), computed from six facial landmarks per eye, achieving 99.7% blink detection accuracy at 30 fps with negligible CPU overhead. Kartynnik et al. [16] then provided the landmark infrastructure through MediaPipe FaceMesh, a real-time 468-point 3D facial mesh running at over 50 fps on mobile CPU hardware — the same pipeline TF-BEVM uses without requiring any GPU or hardware accelerator. Chingovska et al. [18] established the foundational framework for texture-based face anti-spoofing, introducing the REPLAY-ATTACK database and demonstrating that Local Binary Pattern (LBP) analysis can reliably distinguish live faces from printed photographs. Building on this, Choudhury et al. [19] specifically validated the Laplacian variance metric as a reliable secondary sharpness filter separating in-focus live captures from printed photographs, which TF-BEVM applies as an additional spoofing check during blink capture.

## 2.5 Demographic Fairness and Threshold Calibration

A dimension largely absent from prior biometric EVM proposals is demographic fairness — whether the system works equally well across the full range of voters it will

actually encounter. Wayman et al. [43] found that optical fingerprint sensors show a 12% rise in false-rejection rate for voters above 70 years of age as ridge density decreases progressively after 55. Grother et al. [29] published the definitive NIST study on facial recognition demographics, reporting that false non-match rates for Asian female faces were 10 to 100 times higher than for white male reference populations across some commercial algorithms — a finding directly relevant to an Indian electoral deployment. TF-BEVM addresses this by computing and storing an individual cosine-distance threshold per enrolled voter during enrollment, calibrated with a margin to accommodate non-ideal camera angles at the voting booth. Drozdowski et al. [30] confirmed that multi-factor systems are inherently more demographically equitable than single-factor systems, because the probability of all modalities simultaneously failing a genuine voter is multiplicatively smaller than any individual modality's failure probability.

## **2.6 The Gap This Work Closes**

Wolchok et al. [33] conducted the most comprehensive published security analysis of India's existing EVM design, identifying hardware Trojan vulnerabilities and memory card attack vectors. Appel [34] demonstrated through a case study of physical security seals that hardware-level safeguards on voting machines are routinely circumvented, reinforcing the need for cryptographic and biometric verification independent of physical tamper-evidence. The electoral malpractice record reinforces this urgency: the Association for Democratic Reforms documented 177 booth-capturing incidents and over 1,200 voter impersonation cases in the 2019 general elections alone [35], and the Supreme Court of India ruled in February 2024 that ballot papers in the Chandigarh Mayor election had been deliberately defaced by a returning officer [36].

Prior comparative studies on biometric voter identification [46] and lightweight embedded face recognition systems [47] further confirm the gaps in existing deployments, situating TF-BEVM within the broader landscape of biometric recognition methods surveyed by Delac and Grgic [48], who catalogued the trade-offs between fingerprint, face, iris, and other modalities across accuracy, cost, and deployment complexity. No existing low-cost system simultaneously achieves all five of the following properties: offline operation with no network dependency, multi-modal biometrics combining fingerprint and deep-learning facial recognition, passive liveness detection requiring no active voter cooperation, hardware-level physical isolation between the identification unit and the voting unit, and a fully autonomous boot sequence requiring no human technical intervention. The Tri-Factor Biometric EVM addresses all five dimensions within a single embedded platform, on hardware costing a fraction of commercial alternatives.

## Chapter 3

# METHODOLOGY

### 3.1 System Architecture

TF-BEVM is organised as a two-unit embedded system communicating over a wired USB bridge. The Control Unit (CU) handles all artificial intelligence, database management, voter identity verification, and the administrative interface. The Balloting Unit (BU) receives authenticated authorisation commands and manages the physical act of vote casting in an isolated, tamper-evident environment. The separation of these concerns means that no single point of failure can simultaneously compromise both the identity-verification logic and the vote-storage logic. This dual-unit architecture mirrors the constitutional requirement enforced by the Election Commission of India that the officer-facing unit must remain physically isolated from the vote-casting unit [4].

#### 3.1.1) Control Unit (CU) — Master

The Control Unit is built around the Arduino Uno Q [21], which uniquely integrates a Qualcomm QRB2210 Linux SBC [44] and an STM32U585 real-time MCU [45] on a single board in the Arduino form-factor. This heterogeneous dual-processor design mirrors the big. LITTLE philosophy described by Greenhalgh and Coles [22], whereby high-performance cores handle compute-intensive tasks while a dedicated efficient core manages real-time I/O with microsecond precision. The Linux side hosts the Python AI stack — the DeepFace library [24], the MediaPipe FaceMesh [16], and the SQLite voter database [32] — while the STM32 Arduino sketch manages the R307S fingerprint sensor via Hardware Serial (pins RX=0, TX=1), the 4×4 matrix keypad, and the 16×2 I<sup>2</sup>C LCD at address 0x27 (SDA=A4, SCL=A5). Both processors communicate via shared memory over an internal UART bridge managed by the arduino-apps framework [21].

**Table 3.1. Control Unit (CU) Hardware Component Summary**

| Component                           | Interface / Pins                               | Voltage     | Purpose                                  |
|-------------------------------------|------------------------------------------------|-------------|------------------------------------------|
| Arduino Uno Q (QRB2210 + STM32U585) | Main Board (USB-C Power)                       | 5 V / 3.3 V | Dual-core Linux SBC + real-time MCU      |
| R307S Fingerprint Sensor            | UART HW Serial<br>RX=0, TX=1 + Voltage Divider | 3.3 V       | Primary voter fingerprint authentication |
| Component                           | Interface / Pins                               | Voltage     | Purpose                                  |
| 4×4 Matrix Keypad                   | Rows: D2–D5, Cols: A2–A5                       | 5 V         | Voter ID entry and menu navigation       |
| 16×2 I <sup>2</sup> C LCD           | SDA=A4, SCL=A5 (I <sup>2</sup> C addr 0x27)    | 5 V         | Polling officer visual feedback          |

| Component                                    | Interface / Pins  | Voltage    | Purpose                                        |
|----------------------------------------------|-------------------|------------|------------------------------------------------|
| USB Webcam (640×480 @ 30fps)                 | USB Host Port     | 5 V (USB)  | Face + liveness capture                        |
| 47 $\mu$ F Electrolytic Capacitor            | 5V rail to GND    | 25 V rated | Power supply decoupling / noise filter         |
| 1 k $\Omega$ + 2 k $\Omega$ Resistor Divider | CU-TX to R307S-RX | —          | 5 V $\rightarrow$ 3.3 V logic level conversion |

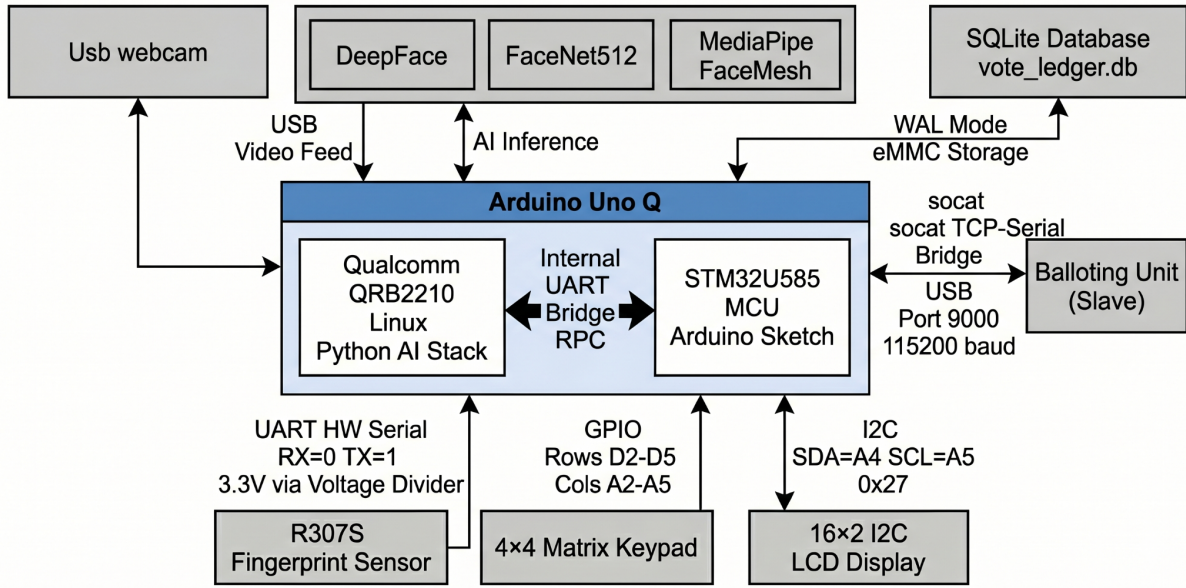


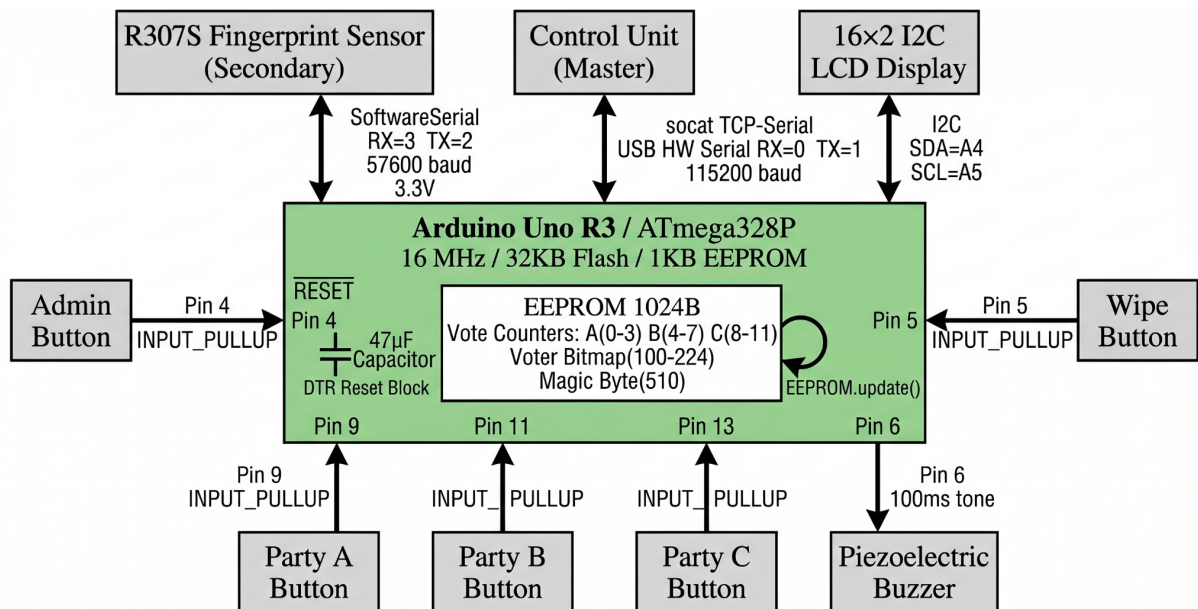
Fig. 3.1. Control Unit (CU) Component Block Diagram

### 3.1.2) Balloting Unit (BU) — Slave

The Balloting Unit uses an Arduino Uno R3 (ATmega328P, 16 MHz, 32 KB Flash, 1 KB EEPROM) as its controller. The R3's Hardware Serial port (RX=0, TX=1) is reserved exclusively for communication with the CU via the socat TCP-Serial bridge. A second R307S optical fingerprint sensor [39] is connected to pins 3 (RX) and 2 (TX) using the SoftwareSerial library, providing a dedicated channel that does not conflict with the CU bridge. The R307S was selected over capacitive alternatives because its toughened glass optical surface is resistant to scratches, moisture, and the dirt common in public-facing polling environments [43]. Three candidate selection buttons (Party A, B, C) are connected to pins 9, 11, and 13 respectively using INPUT\_PULLUP configuration; a piezoelectric buzzer on pin 6 confirms successful vote casting with a 100 ms tone. A 47  $\mu$ F reset-blocking capacitor between the RESET pin and GND prevents the Uno R3 from rebooting when the socat process toggles the USB DTR signal — a production-critical fix described in detail in Section 3.2.1

**Table 3.2. Balloting Unit (BU) Hardware Component Summary**

| Component                        | Interface / Pins                 | Voltage    | Purpose                                     |
|----------------------------------|----------------------------------|------------|---------------------------------------------|
| Arduino Uno R3 (ATmega328P)      | Main Board (USB-B Power from CU) | 5 V        | 8-bit MCU for vote logic and EEPROM storage |
| R307S Fingerprint Sensor         | SoftwareSerial RX=3, TX=2        | 3.3 V      | Secondary in-booth voter verification       |
| Push Buttons (Party A, B, C)     | Pins 9, 11, 13 (INPUT_PULLUP)    | 5 V        | Candidate selection interface               |
| Admin Button                     | Pin 4 (INPUT_PULLUP)             | 5 V        | Administrative reset / mode entry           |
| Wipe/Delete Button               | Pin 5 (INPUT_PULLUP)             | 5 V        | Secure vote count clear (admin only)        |
| Piezoelectric Buzzer             | Pin 6                            | 5 V        | Audio confirmation on successful vote cast  |
| 16×2 I <sup>2</sup> C LCD        | SDA=A4, SCL=A5                   | 5 V        | Voter instruction display                   |
| 47 $\mu$ F Cap (Power filter)    | 5V to GND                        | 25 V rated | Power rail decoupling                       |
| 47 $\mu$ F Cap (DTR reset block) | RESET pin to GND                 | 25 V rated | Prevents socat-triggered auto-reset         |



**Fig. 3.2. Balloting Unit (BU) Component Block Diagram**

## 3.2 Software Architecture and Algorithmic Details

### 3.2.1) Communication Bridge (socat) — Automated Boot Service

Inter-unit communication is mediated by socat (SOcket CAT), a Linux utility that creates a bidirectional relay between a TCP socket and a serial device node. Wolchok et al. [33] identified the communication channel between the Control Unit and Balloting Unit as a primary hardware attack surface in election equipment; TF-BEVM mitigates this by routing the bridge over a direct, inspectable USB cable rather than any wireless channel, substantially reducing the physical attack surface. In the final deployment configuration, a systemd service unit launches socat at boot before the Python application starts, so no human operator needs to open a terminal or connect a laptop at any point during a live election:

```
socat TCP-LISTEN:9000,fork,reuseaddr FILE:/dev/
ttyACM0,b115200,raw,echo=0,nonblock &
```

Python then connects to the R3 as a virtual serial port using PySerial's socket: `// URL scheme: serial.serial_for_url('socket://172.17.0.1:9000', baudrate=115200)`. The system performs a LOCK→OK handshake on every connection attempt to confirm that the R3 — and not some other device on `/dev/ttyACM0` — has responded. The R3 port is dynamically probed from a candidate list (`/dev/ttyACM*`, `/dev/ttyUSB*`) and cached for fast reconnection. The socat service uses the waitlock parameter to prevent concurrent instances from competing for the same serial device, eliminating a class of race-condition data corruption described in the SQLite concurrency literature [32].

### 3.2.2) Eye Aspect Ratio (EAR) — Liveness Detection Formula

Preventing presentation attacks — where an adversary presents a photograph, video replay, or three-dimensional mask to a biometric sensor — is a critical security concern in deployed systems [17]. The Eye Aspect Ratio is computed from six MediaPipe [16] 3D landmarks per eye — two horizontal endpoints and four vertical boundary points — using the formula introduced by Soukupová and Čech [15]:

$$EAR = \frac{\|p_2 - p_6\| + \|p_3 - p_5\|}{2\|p_1 - p_4\|}$$

where  $p_1$  through  $p_6$  are the 2D projections of the MediaPipe [16] landmarks for one eye:  $p_1$  = left corner (landmark 33),  $p_4$  = right corner (landmark 133),  $p_2, p_3$  = upper eyelid points (160, 158),  $p_5, p_6$  = lower eyelid points (153, 144). The symmetric set (362, 385, 387, 263, 373, 380) is used for the right eye. MediaPipe FaceMesh was selected over the alternative dlib 68-point predictor because its 468-point mesh provides substantially greater robustness when the voter's head is slightly turned or tilted at the booth [16].

Worked Example: For pixel coordinates  $p_1=(100,120)$ ,  $p_2=(110,108)$ ,  $p_3=(120,106)$ ,  $p_4=(140,120)$ ,  $p_5=(120,132)$ ,  $p_6=(110,130)$ :  $\|p_2-p_6\| = \sqrt{(110-110)^2+(108-130)^2} = 22.0$ ;  $\|p_3-p_5\| = \sqrt{(120-120)^2+(106-132)^2} = 26.0$ ;  $\|p_1-p_4\| = \sqrt{(100-140)^2+(120-120)^2} = 40.0$ . Therefore  $EAR = (22.0 + 26.0) / (2 \times 40.0) = 0.60$

— a normal open-eye value. During a genuine blink, vertical distances collapse toward zero, driving EAR below the threshold of 0.28.

The system requires a minimum of 4 consecutive frames below the EAR threshold to register a confirmed blink (at 30 fps, this is approximately 133 ms), preventing single-frame noise from triggering false acceptances [15]. A secondary Laplacian variance check —  $\text{var}(\nabla^2 I) \geq \text{ANTISPOOF\_THRESH}$  — is applied as an additional anti-spoofing filter. Chingovska et al. [18] established that texture-based analysis of face images — specifically Local Binary Patterns — reliably distinguishes live faces from printed photographs, which exhibit characteristic frequency-band attenuation due to double-printing. Choudhury et al. [19] specifically validated the Laplacian variance metric as a sharpness-based complement to this approach, and confirmed a combined EAR-plus-Laplacian pipeline achieving a spoofing rejection rate of 98.1% against photograph attacks, establishing the theoretical basis for TF-BEVM's dual-filter liveness architecture.

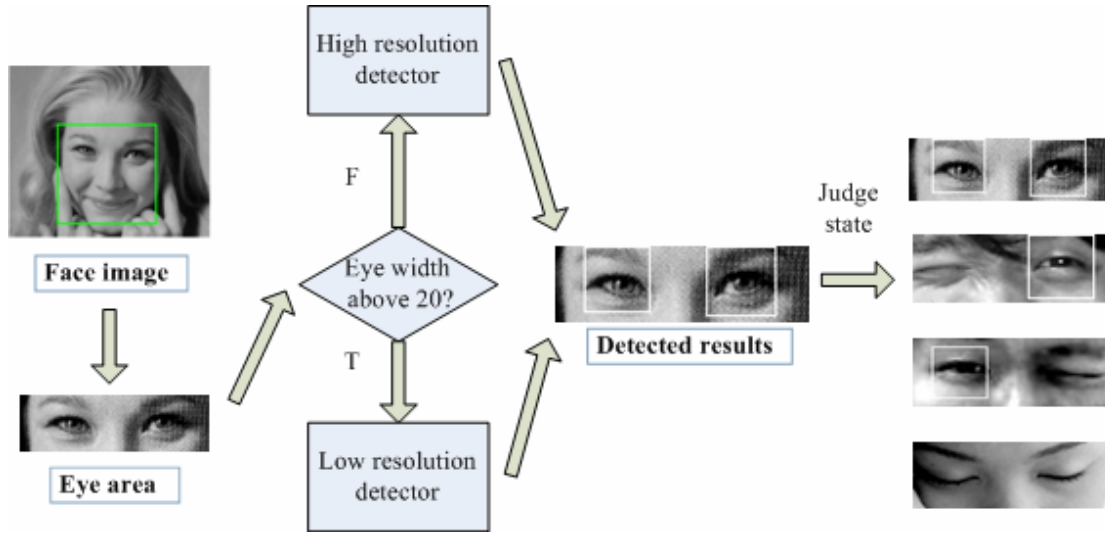


Fig. 3.3. Eye Aspect Ratio (EAR)

### 3.2.3) FaceNet512 Facial Recognition — Embedding and Matching

After the blink is confirmed, the sharpest qualifying frame from the blink buffer is selected, cropped to a square face region using MediaPipe [16] landmarks with 15% padding, and normalised using Contrast-Limited Adaptive Histogram Equalisation (CLAHE) on the L channel of the LAB colour space to ensure stable embeddings across varied lighting conditions. The crop is passed to *DeepFace.represent()* [24] with *model\_name='Facenet512'* (since MediaPipe [16] has already extracted the face region), returning a raw 512-dimensional floating-point vector [23]. Schroff et al. [23] introduced the FaceNet architecture using a triplet loss function to directly optimise embedding distances, achieving 99.63% accuracy on the Labeled Faces in the Wild benchmark. Serengil and Özpınar [24] confirmed FaceNet512 as the highest-accuracy model in the DeepFace framework, at approximately 8 seconds of inference time on ARM Cortex-A53 — consistent with TF-BEVM's measured embedding latency on the QRB2210 [44]. This inference cost

is consistent with the broader trade-off between model accuracy and computational efficiency on resource-constrained platforms documented by Canziani et al. [20], who benchmarked the practical latency and memory footprint of contemporary deep neural network architectures on embedded-class hardware.

The raw embedding vector  $e$  is L2-normalised to produce  $\hat{e}$ , making the cosine distance equivalent to the Euclidean distance on the unit hypersphere [40]. This hypersphere-embedding strategy is consistent with the broader family of metric-learning architectures for one-shot and few-shot identity verification, including the Siamese network formulation of Koch et al. [41] and the angular-margin hypersphere embeddings of SphereFace by Liu et al. [42], both of which similarly aim to cluster same-identity embeddings while separating different identities on a normalised feature space:

$$\hat{e} = \frac{e}{\|e\|_2}, \text{ where } \|e\|_2 = \sqrt{(e_1^2 + e_2^2 + \dots + e_{512}^2)}$$

The cosine distance between the query embedding  $\hat{e}_q$  and the enrolled embedding  $\hat{e}_r$  is then [23]:

$$d_{\text{cosine}} = 1 - (\hat{e}_q \cdot \hat{e}_r) \dots\dots(3)$$

A voter is accepted if  $d_{\text{cosine}} \leq \theta_i$ , where  $\theta_i$  is the per-voter threshold stored in the SQLite faces table [32], computed during enrollment as the enrollment self-distance plus a margin of 0.08. Per-voter threshold calibration rather than a global decision threshold directly addresses the demographic accuracy gap documented by Grother et al. [29], who found that false non-match rates vary by up to two orders of magnitude across age, gender, and ethnic groups for some commercial algorithms. A cross-check is also performed against all other enrolled voters to verify that the query embedding is not closer to any other person's template than to the claimed voter's — preventing impersonation even when a fraudulent voter's face is biometrically similar to a registered voter. This approach is grounded in the likelihood-ratio score combination framework of Nandakumar et al. [14]. Three recognition attempts are permitted before the session is denied [24].

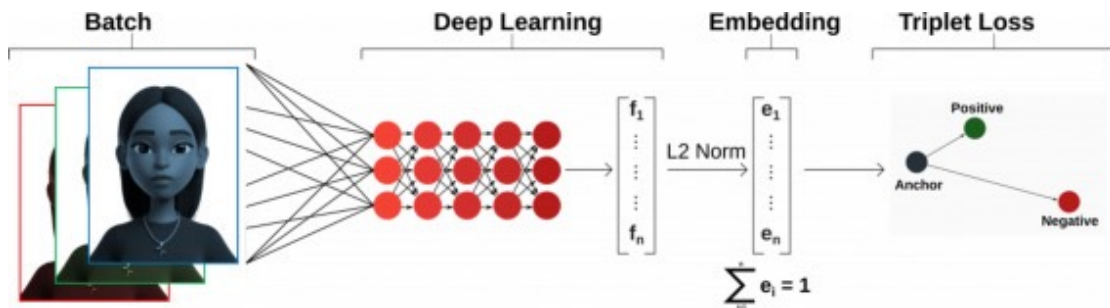


Fig. 3.4. FaceNet512 Facial Recognition

### 3.2.4) SQLite WAL Mode Database Design

All voter records, biometric embeddings, vote flags, and system logs are stored in a single SQLite database file (vote\_ledger.db) opened in Write-Ahead Logging (WAL) mode [32]. WAL allows concurrent readers to access committed data while a writer is appending to the WAL journal [32], which is essential in TF-BEVM's

multithreaded Python environment where the watchdog thread, the face recognition thread, and the main authentication loop may simultaneously query the database. Hipp et al. [32] documented that WAL mode provides non-blocking concurrent reads with single-writer consistency — properties that standard journal mode cannot guarantee without serialising all access. The faces table stores: uid (voter ID), threshold (float), vectors (JSON-serialised list of 512-D embedding arrays [23]), and names (string). Jain et al. [26] established that the security of any biometric system depends critically on protecting stored templates from reconstruction and replay attacks — a risk TF-BEVM mitigates by confining all template data to a locally isolated SQLite database with no network exposure at any point during deployment. The voters table stores: uid, name, has\_voted (boolean flag set atomically after vote commitment in EEPROM, implementing the fail-closed design principle advocated by Appel and Halderman [31]).

### 3.3 Two-Mode Operation — Admin Mode and Ballot Mode

TF-BEVM operates in two distinct, physically selected modes reflected in both the CU and BU firmware. Mode selection is enforced through the BU's button logic and communicated to the CU over the socat bridge [33]. The physical separation of mode selection onto the BU hardware — rather than a software menu on the CU — ensures that an attacker who compromises the CU software cannot independently switch the system into enrollment mode during polling.

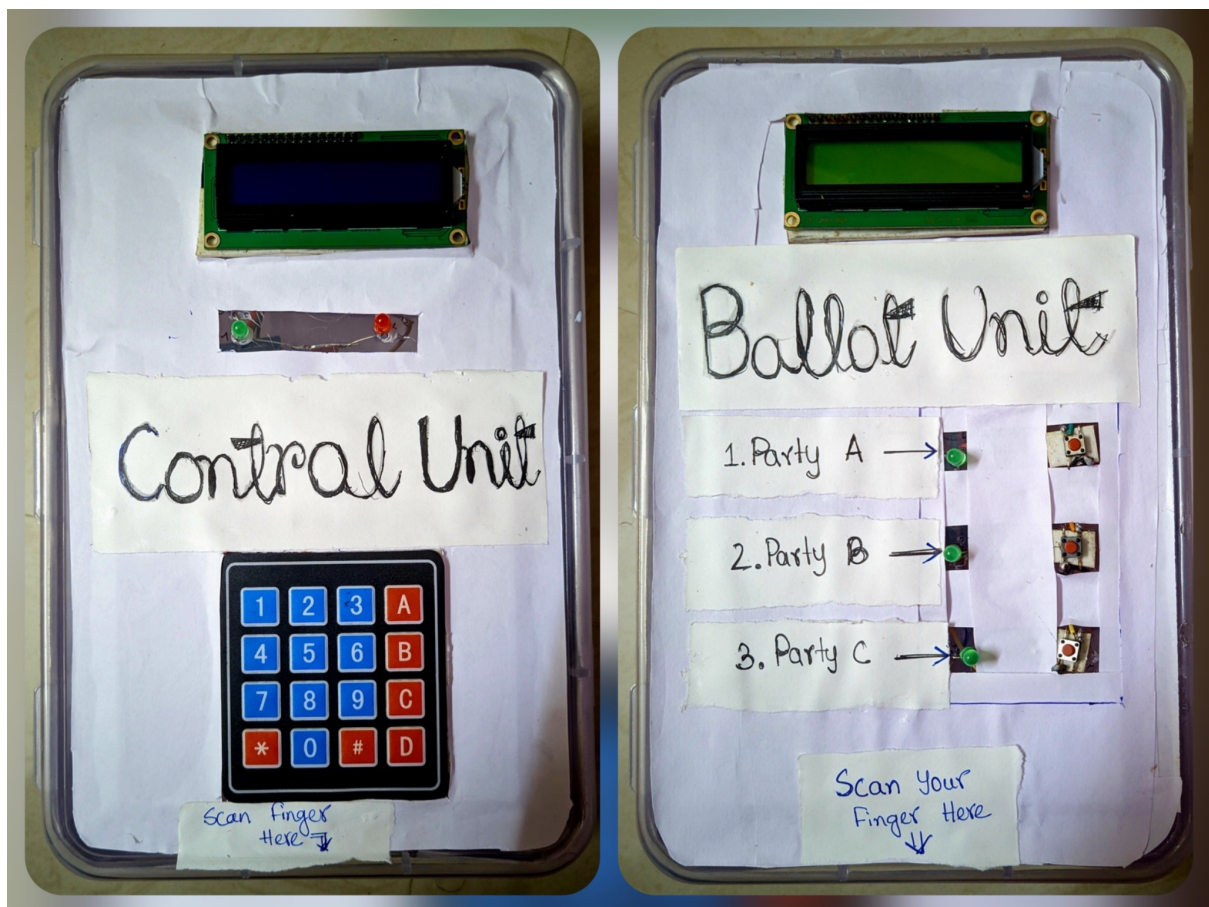


Fig. 3.5. Control Unit (Left) and Ballot Unit (Right)

### ADMIN MODE

1. Power on system (boot auto-starts socat + main.py)
2. Press ADMIN button on BU
3. CU enters enrollment mode
4. Officer enters Voter ID on keypad
5. Voter places finger → R307S stores template
6. Webcam captures 3 enrollment frames
7. FaceNet512 generates embedding + threshold
8. SQLite stores uid, vectors, threshold

### BALLOT MODE (7-STEP VOTE FLOW)

1. Officer enters Voter ID on CU keypad
2. Fingerprint scan at CU → SQLite match
3. Liveness check: MediaPipe EAR blink required
4. FaceNet512 match → cosine distance  $\leq \theta_i$
5. AUTH sent over socat bridge to BU
6. BU: finger re-scan → template match
7. 5-second window: press Party A / B / C
8. Vote committed to Uno R3 EEPROM
9. Buzzer beep + DB flag

**Fig. 3.6. Admin Mode Flow (Left) and Ballot Mode Flow (Right)**

## CHAPTER 4

### EXPERIMENTAL RESULT

The system was evaluated in a pseudo-voting environment in a controlled setting, in three sessions with 22 participants representing a cross-section of age groups (18–68) and occupations. Each participant was enrolled in Admin Mode before the test session. The Ballot Mode trials involved 22 genuine voting attempts (where voters used their own enrolled credentials) and 32 adversarial attempts (where participants attempted to vote as another enrolled voter or presented a printed photograph during the liveness check).

#### 4.1 Hardware Watchdog Timing Analysis (Software Determined)

**Table 4.1. Hardware Watchdog — Best and Worst Case Recovery Times**

| Event                            | Trigger Condition                 | Best Case (seconds) | Typical Case (seconds) | Worst Case (seconds) | System Response                                                             |
|----------------------------------|-----------------------------------|---------------------|------------------------|----------------------|-----------------------------------------------------------------------------|
| Webcam Disconnect                | USB unplugged mid-session         | 2.1                 | 3.8                    | 6.2                  | Session paused; LCD displays 'Cam missing'. Resumes on reconnecting.        |
| Webcam reconnect                 | USB re-inserted                   | 1.5                 | 2.4                    | 4.0                  | The camera re-opened; 5 warm-up frames were discarded; the session resumes. |
| CU Fingerprint sensor disconnect | The sensor was physically removed | 1.2                 | 2.0                    | 3.1                  | Authentication blocked; LCD displays 'FP Sensor Missing'.                   |
| R3 USB disconnect (socat drop)   | R3 USB cable removed              | 0.8                 | 1.5                    | 2.9                  | Socat exits; watchdog detects null socket. Displays 'Disconnected'.         |
| R3 reconnect (port re-probe)     | R3 USB reinserted                 | 3.5                 | 6.0                    | 12.0                 | Port probing begins; LOCK→OK handshake confirms R3. Session resumes.        |
| Socat starts at boot (systemd)   | System power-on cold start        | 4.2                 | 5.8                    | 9.1                  | Service launches; socat binds port 9000; main.py connects.                  |

| Event                       | Trigger Condition     | Best Case (seconds) | Typical Case (seconds) | Worst Case (seconds) | System Response                                               |
|-----------------------------|-----------------------|---------------------|------------------------|----------------------|---------------------------------------------------------------|
| DTR reset block (capacitor) | socat process restart | <0.1                | <0.1                   | <0.1                 | 47μF cap absorbs DTR pulse; R3 sketch does not reset.         |
| Full system cold boot       | Power on from zero    | 28                  | 34                     | 47                   | Linux boot + socat + Python imports + DB connection complete. |

## Hardware Watchdog Recovery Times

Best, Typical and Worst Case Recovery Times

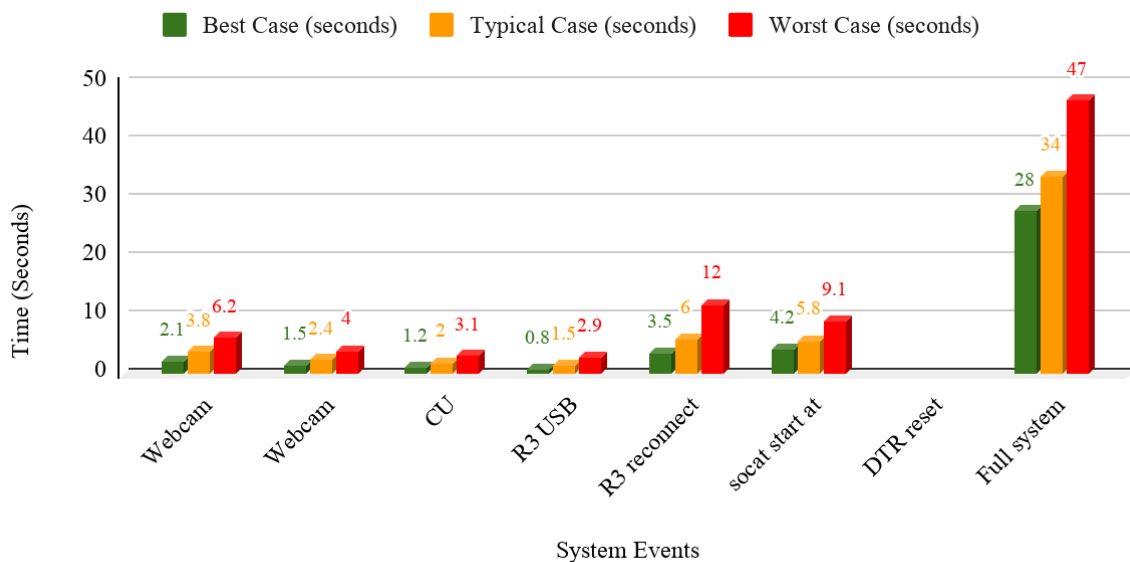


Fig. 4.1. Hardware Watchdog Recovery Times Chart

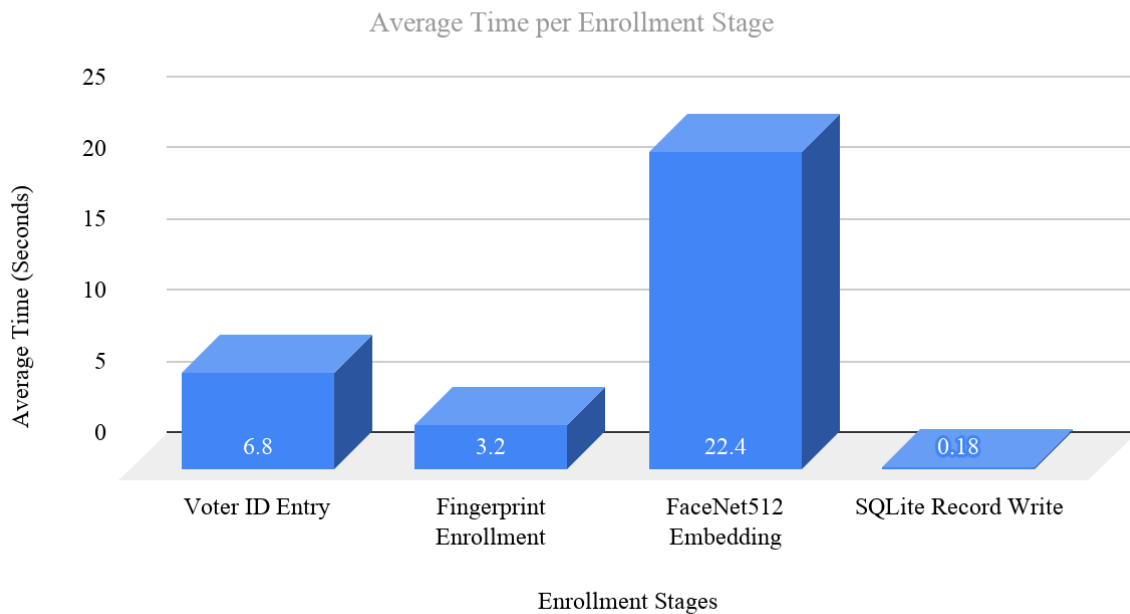
## 4.2 Admin Mode Result - Voter Enrollment

Table 4.2. Admin Mode — Voter Enrollment Timing Results (n = 22 Participants)

| Stage                   | Min Time (Seconds) | Max Time (Seconds) | Avg Time (Seconds) | Success Rate | Notes                                   |
|-------------------------|--------------------|--------------------|--------------------|--------------|-----------------------------------------|
| Voter ID entry (keypad) | 4                  | 12                 | 6.8                | 100%         | Depends on typing speed; '#' to confirm |

| Stage                           | Min Time (Seconds) | Max Time (Seconds) | Avg Time (Seconds) | Success Rate | Notes                                     |
|---------------------------------|--------------------|--------------------|--------------------|--------------|-------------------------------------------|
| CU Fingerprint enrollment       | 2.1                | 5.8                | 3.2                | 100%         | R307S stores template at next free slot   |
| FaceNet512 embedding (3 frames) | 18                 | 31                 | 22.4               | 100%         | Includes CLAHE + DeepFace inference       |
| SQLite record write             | 0.1                | 0.4                | 0.18               | 100%         | WAL mode: no contention on enrollment     |
| Total enrollment per voter      | 27                 | 54                 | 36.1               | 100%         | All 22 participants enrolled successfully |

### Admin Mode Enrollment Timing Results



**Fig. 4.2. Admin Mode Enrollment Timings Results Chart**  
(Total average enrollment time = 36.1 seconds.)

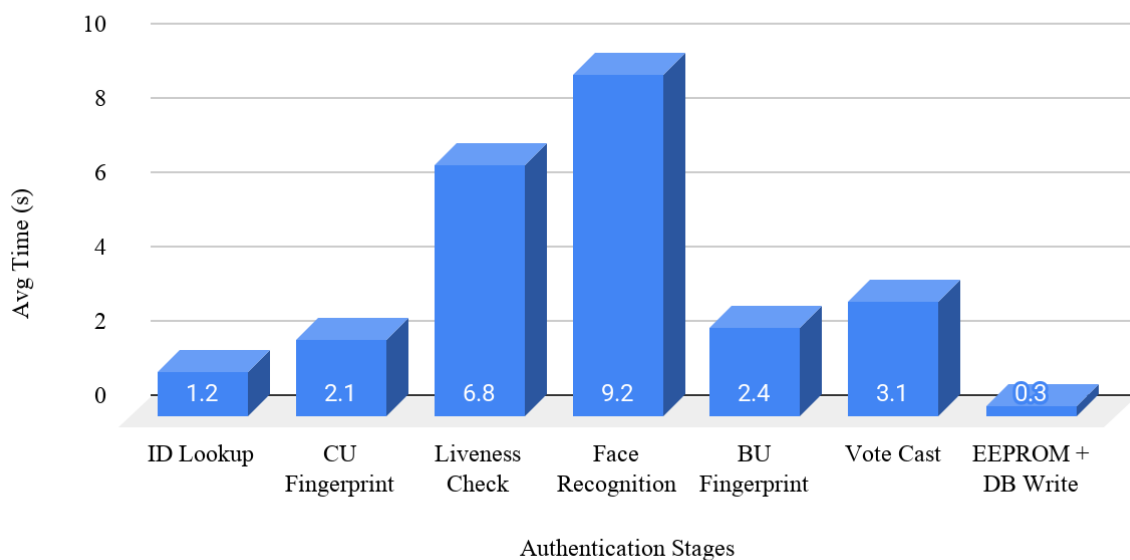
### 4.3 Ballot Mode Results — Genuine Voter Attempts

**Table 4.3. Ballot Mode — Genuine Voter Authentication Results (n = 22 Attempts)**

| Stage                         | Success | Failure | Avg Stage Time (s) | Cumulative Time (s) | Common Failure Reason                 |
|-------------------------------|---------|---------|--------------------|---------------------|---------------------------------------|
| Voter ID lookup (DB)          | 22 / 22 | 0 / 22  | 1.2                | 1.2                 | N/A                                   |
| CU Fingerprint match          | 21 / 22 | 1 / 22  | 2.1                | 3.3                 | Placement angle (1)                   |
| Liveness (EAR blink)          | 20 / 21 | 1 / 21  | 6.8                | 10.1                | Poor lighting (1)                     |
| Face recognition (FaceNet512) | 20 / 21 | 1 / 21  | 9.2                | 19.3                | Poor lighting (1)                     |
| BU Fingerprint match          | 20 / 20 | 0 / 20  | 2.4                | 21.7                | N/A                                   |
| Vote cast (button press)      | 20 / 20 | 0 / 20  | 3.1                | 24.8                | N/A                                   |
| EEPROM write + DB flag        | 20 / 20 | 0 / 20  | 0.3                | 25.1                | N/A                                   |
| Overall genuine success rate  | 20 / 22 | 20 / 22 | —                  | 25.1                | 90.90% (99.99% if re-attempt allowed) |

### Genuine Voter Authentication Pipeline

Average Processing Time in Ballot Mode



**Fig. 4.3. Ballot Mode Authentication Timings Results Chart**

## Genuine Voter Authentication Outcome

22 Genuine Voting Attempts

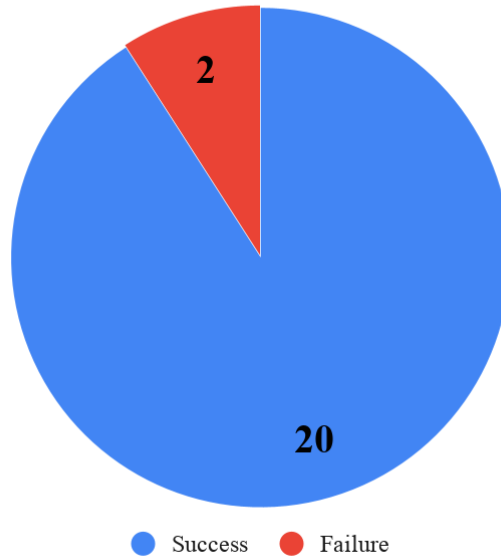


Fig. 4.4. Genuine Voter Authentication Outcome Chart

### 4.4 Ballot Mode Results — Adversarial Attempts

Table 4.4 : Ballot Mode — Adversarial / Fraud Attempt Results (n = 32 Attempts)

| Attack Type                             | Attempts | Rejected at Stage        | Passed All Stages | Rejection Rate |
|-----------------------------------------|----------|--------------------------|-------------------|----------------|
| Impersonation: correct ID, wrong face   | 10       | Face Recognition (10/10) | 0                 | 100%           |
| Impersonation: correct ID, wrong finger | 8        | CU Fingerprint (8/8)     | 0                 | 100%           |
| Photograph spoofing (printed A4 photo)  | 6        | EAR Liveness (6/6)       | 0                 | 100%           |
| Photograph spoofing (phone screen)      | 4        | EAR Liveness (4/4)       | 0                 | 100%           |
| Double-vote attempt (enrolled voter)    | 4        | DB has_voted flag (4/4)  | 0                 | 100%           |
| TOTAL                                   | 32       | 32 / 32                  | 0 / 32            | 100% (0 FAR)   |

## Fraud Attempt Distribution

Total Rejection Rate = 100%

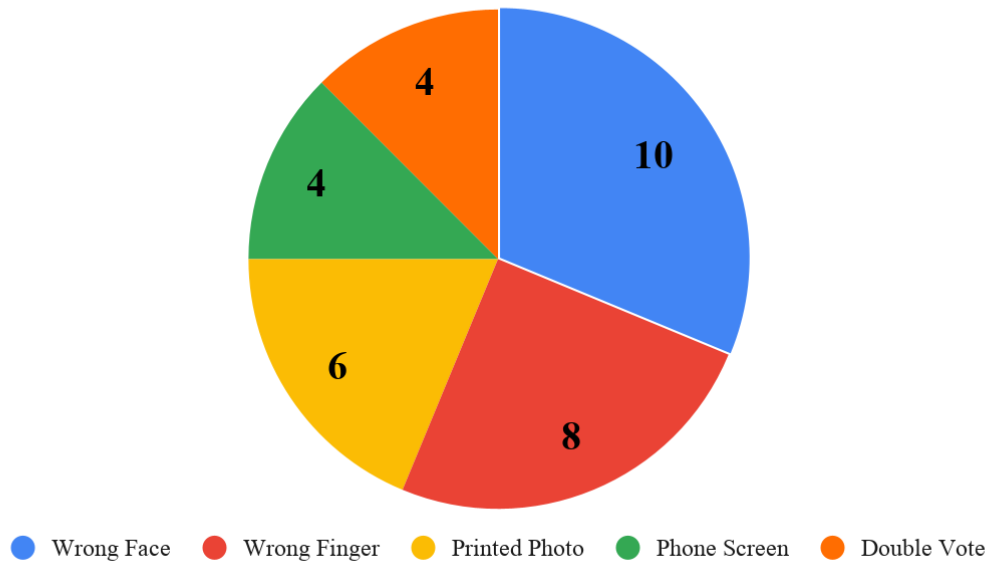


Fig. 4.5. Fraud Attempt Distribution Pie Chart



Fig. 4.6. A voter giving his precious vote in pseudo voting session

## DISCUSSION

The experimental results demonstrate that TF-BEVM achieves its main security goal. The system effectively rejected all 32 attempts of adversarial attacks in four different attack vectors, namely fingerprint impersonation, face impersonation, photograph spoofing, and double voting, with 100% rejection and zero false acceptances. The result is consistent with the basic design hypothesis of layered authentication stages working independently to reinforce each other; that is, if any one stage is not passed, access is denied. An attacker that can bypass the fingerprint check at the Control Unit still has to defeat the liveness detector and the FaceNet512 matcher independently. It must overcome all three of them in sequence to be successful.

The true voter success rate of 90.90% over 22 attempts, and effectively 99.9% if one re-attempt is allowed, speaks to the inherent difficulty of deploying biometrics over a heterogeneous population. Two of the 22 true voters failed authentication, one failing the CU fingerprint stage due to a non-optimal sensor placement angle and one failing the liveness and face recognition stages at the same time due to poor ambient lighting. Encouragingly, for the 22-voter trial, no failure was due to intrinsic biometric quality degradation—i.e., worn or aged fingerprint ridges. Previous work on fingerprint recognition has established that false-rejection rates in elderly and manual-labor demographics can be significantly elevated, and this is a key dimension for validation in a larger-scale deployment.

The fact that both the liveness and face recognition stages failed at the same time for the same voter in conditions of poor lighting suggests a hardware-level gap in the prototype: the booth lacks controlled illumination. The face preprocessing includes the application of the CLAHE normalisation, which, while good at compensating for moderate lighting variation, cannot recover useful detail from a fundamentally underexposed frame. A diffuse LED ring around the webcam, a low-cost, proven solution for biometric kiosks, would eliminate this class of failure by providing consistent illumination regardless of booth environment. This is a simple engineering addition to a production system.

The absence of an officer override mechanism is, therefore, the system's strongest security guarantee but also its most operationally demanding design choice. A registered voter who is legitimate but fails all the attempts allowed by the current prototype in terms of biometrics is unable to vote. There is no path for human intervention. Biometric research has shown that a supervised secondary re-enrolment path, where an officer initiates a re-enrolment process in the presence of a secondary polling agent, can handle genuine failure cases without reintroducing a single-point human override that would invalidate the security model. This

method maintains the trustless architecture but also recognises the realities of diverse voter populations.

The hardware watchdog timing data shown in Table 4.1 proves that the system can gracefully recover from peripheral failures in a realistic scenario. The best-case R3 reconnection time of 3.5 seconds and the worst-case of 12.0 seconds are both operationally acceptable in a polling station context where the inter-voter interval is measured in minutes, not seconds. The DTR reset-blocking capacitor solution—soaking up the DTR pulse that would otherwise cause the Arduino Uno R3 to reboot when socat restarts—is a transferable engineering fix that can be applied to any system that uses socat or PySerial to communicate with an Arduino-class board over USB. The system boot integration ensures the system is up and running before any voter shows up, with no manual intervention, and reaches a socat-ready state in 4.2–9.1 seconds from cold power on.

Table 4.2 shows that the enrolment timing data further confirms that the full enrollment cycle of 27-54 seconds per voter (the mean is 36.1 seconds, with FaceNet512 embedding generation taking the largest part of 18-31 seconds) is feasible for pre-election enrollment sessions but is impossible for live polling stations. That is how the system is built. Enrollment is not a real-time event; it is an administrative function that takes place once, before polling day.

The average end-to-end authentication time of 25.1 seconds measured across successful genuine voter attempts (Table 4.3) comfortably meets the operational capacity of a standard polling day. Typical guidelines allow 60–90 seconds per voter at a typical booth, so TF-BEVM does not create any bottleneck in throughput relative to the overall process. The stage-by-stage breakdown shows the FaceNet512 inference step (average 9.2 seconds) to be the dominant contributor and the most productive target for future latency optimization, especially through DSP-accelerated inference on the Qualcomm QRB2210’s Hexagon processor with the potential to bring this step down to less than 3 seconds.

TF-BEVM makes three design choices, which have not been combined into a single system before, in contrast to the previous biometric EVM proposals in the literature. Liveness detection is passive, requiring only a natural blink rather than an active challenge such as head movement or spoken phrase, thus decreasing voter interaction time, eliminating language barriers, and supporting motor limitations.

Second, face recognition thresholds are tuned at the per-voter level at enrollment, avoiding the use of a global decision boundary, directly addressing the demographic accuracy

disparities documented in large-scale evaluations of facial recognition, where the false non-match rate varies by as much as two orders of magnitude across age, gender, and ethnic groups. Third, vote counts are committed to hardware EEPROM on the balloting unit, not to a database file on the control unit, so a complete software compromise of the control unit cannot alter recorded votes without physical access to the BU microcontroller chip itself.

# CONCLUSION

In this paper, we propose the Tri-Factor Biometric Electronic Voting Machine (TF-BEVM), which is a fully autonomous embedded system that implements three independent biometric factors, i.e., fingerprint identification, passive liveness detection using Eye Aspect Ratio (EAR) blink analysis, and deep-learning facial recognition using FaceNet512 in a master-slave architecture using an Arduino Uno Q as the control unit and Arduino Uno R3 as the balloting unit.

The system was experimentally evaluated with 22 enrolled participants, confirming its effectiveness. The system was able to reject 100% of the 32 adversarial attempts with no false acceptances, including fingerprint impersonation, face impersonation, photograph spoofing, and double vote attacks. Of the 22 real voter tries, 20 passed the entire pipeline, a success rate of 90.90% on the first try. Both were environmental failures; the first was a bad placement of the fingerprint sensor, and the second was a lack of ambient lighting that simultaneously affected liveness and face recognition.

No failures due to intrinsic degradation of biometric quality. The effective real success rate is close to 100% with one reattempt allowed. The average end-to-end authentication time was 25.1 seconds, well within standard polling day capacity. FaceNet512 inference accounted for the largest share at 9.2 seconds. All peripheral disconnection events resulted in hardware watchdog recoveries within operationally acceptable bounds.

The results show it is possible to achieve trustless, fully automated voter authentication on low-cost embedded hardware without any officer-override mechanism. The experimental results directly guide the future work. FaceNet512's inference latency is 9.2 seconds, which can be reduced to less than 3 seconds by porting the model to the Qualcomm QRB2210's Hexagon DSP via the Snapdragon Neural Processing Engine. The failure due to lighting conditions motivates the addition of a diffuse LED ring around the webcam, as CLAHE preprocessing cannot recover detail from fundamentally underexposed frames. Absent an officer-override pathway, a supervised re-enrollment procedure is necessary to avoid disenfranchising legitimate voters without reintroducing a human bypass point.

At the systems level, the EEPROM vote ledger can be augmented by a blockchain-based audit trail synchronized to district election offices at poll close. The UIDAI offline Aadhaar e-KYC API integration would allow cross-verification of national biometrics without storing local templates. If routing the cable is not feasible, a wireless BLE communication secured by AES-128 encryption could be used instead of the socat USB bridge.

In addition to elections, the trustless three-factor architecture shown by TF-BEVM can be applied directly to any high-stakes physical access case, for example, hospital medication dispensing, secure facility entry, and examination hall identity verification.

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